



Assnani Dental AI Chatbot

API Reference & Endpoint Documentation

Generated: June 07, 2026 at 18:11

Version: 1.0.0

Framework: FastAPI (Python)

Base URL: <http://localhost:7860> | HF Spaces Port 7860

Total Endpoints: 7

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ENDPOINT 1 OF 7

GET /

Serve Main Chatbot Page

Returns the static index.html file that contains the full chatbot UI (HTML + CSS + JS). This is the entry point for the web application.

EXAMPLE RESPONSE

```
<!DOCTYPE html><html>...full chatbot UI...</html>
```

ENDPOINT 2 OF 7

GET

/health

Health Check

Simple health check endpoint to verify the API is running. Used by monitoring tools and Hugging Face Spaces to check service availability.

RESPONSE BODY (JSON)

Key	Type	Example	Description
status	string	healthy	Service status indicator
service	string	Assnani Dental Chatbot	Service name identifier

EXAMPLE RESPONSE

```
{
  "status": "healthy",
  "service": "Assnani Dental Chatbot"
}
```

ENDPOINT 3 OF 7

POST

/api/analyze-symptoms

Analyze Patient Symptoms

Receives patient symptom data from the chatbot interview and runs the weighted scoring algorithm to produce a risk assessment (low / moderate / high) with recommendations and home-care tips.

REQUEST BODY (SymptomData)

Key	Type	Example	Description
has_pain	boolean	True	Whether the patient reports dental pain
pain_location	string	lower-left	Quadrant/area of pain (e.g. upper-right, lower-left)
pain_type	string	throbbing	Type of pain: throbbing sharp dull sensitivity
pain_intensity	integer	7	Pain severity on a 0-10 scale (clamped)
pain_duration	string	3-7 days	How long pain has lasted: today 1-3 days 3-7 days 1+ week
pain_triggers	list[str]	["hot", "cold", "biting", "sponta...]	What triggers the pain
has_swelling	boolean	True	Whether swelling is present
swelling_severity	string	severe	Severity: mild moderate severe
has_fever	boolean	False	Whether the patient has a fever
difficulty_opening	boolean	False	Whether the patient has trismus (difficulty opening mouth)
has_trauma	boolean	False	History of dental trauma / injury
has_broken_tooth	boolean	True	Whether a tooth is visibly broken or chipped
previous_root_canal	boolean	False	Whether the affected tooth had a previous root canal
last_visit	string	1+ year	Last dental visit: < 6 months 6-12 months 1+ year never
recent_extraction	boolean	False	Whether the patient had a recent tooth extraction

EXAMPLE REQUEST

```
{
  "has_pain": true,
  "pain_location": "lower-left",
  "pain_type": "throbbing",
  "pain_intensity": 7,
```

```
{
  "pain_duration": "3-7 days",
  "pain_triggers": [
    "hot",
    "cold",
    "biting"
  ],
  "has_swelling": true,
  "swelling_severity": "severe",
  "has_fever": false,
  "difficulty_opening": false,
  "has_trauma": false,
  "has_broken_tooth": true,
  "previous_root_canal": false,
  "last_visit": "1+ year",
  "recent_extraction": false
}
```

RESPONSE BODY (JSON)

Key	Type	Example	Description
score	integer	72	Total weighted symptom score
risk_level	string	high	Risk level: low moderate high
risk_emoji	string	🟡	Colour-coded emoji for the risk level
recommendation	string	Based on your symptoms, an X-r...	Human-readable recommendation text
factors	list[tuple]	[["Dental pain reported", 10],...	List of [description, score] factor tuples
home_care	list[str]	["Take Ibuprofen 400mg as need...	Personalised home-care tips
xray_recommended	boolean	True	Whether an X-ray is recommended based on risk

EXAMPLE RESPONSE

```
{
  "score": 72,
  "risk_level": "high",
  "risk_emoji": "🟡",
  "recommendation": "Based on your symptoms, an X-ray examination is strongly recommended as soon as possible.",
  "factors": [
    [
      "Dental pain reported",
      10
    ],
    [
      "Throbbing pain (possible pulpitis)",
      20
    ],
    [
      "Spontaneous pain (possible nerve involvement)",
      30
    ]
  ]
}
```

```
    ],  
    [  
      "High pain intensity (7/10)",  
      25  
    ]  
  ],  
  "home_care": [  
    "Take over-the-counter pain relief (Ibuprofen 400mg) as needed.",  
    "Avoid very hot or cold foods/drinks on the affected side.",  
    "Do not delay seeking professional dental care."  
  ],  
  "xray_recommended": true  
}
```

ENDPOINT 4 OF 7

POST /api/detect-xray

Detect Dental Conditions from X-ray

Accepts dental X-ray image(s) or PDF uploads, extracts images from PDFs via PyMuPDF, forwards each to the external YOLOv8 detection API with retry logic, and returns combined detection results with annotated images.

REQUEST BODY (File Upload)

Key	Type	Example	Description
images	List[UploadFile]	<binary file data>	One or more image files (PNG, JPG, JPEG, WebP) or PDF files containing dental X-rays

EXAMPLE REQUEST

```
# Using cURL:
curl -X POST http://localhost:7860/api/detect-xray \
  -F "images=@dental_xray.jpg"

# Using Python (requests):
import requests

url = "http://localhost:7860/api/detect-xray"
files = {"images": open("dental_xray.jpg", "rb")}
response = requests.post(url, files=files)
print(response.json())

# Multiple files:
files = [
    ("images", open("xray1.jpg", "rb")),
    ("images", open("xray2.jpg", "rb"))
]
response = requests.post(url, files=files)
```

RESPONSE BODY (JSON)

Key	Type	Example	Description
results	list[dict]	[{...result per image...}]	Array of per-image result objects
results[].detections	list[dict]	[{"class_name": "cavity", "confi...]	List of detected dental conditions with bounding boxes
results[].detections[].class_name	string	cavity	Detected class: cavity filling implant impacted
results[].detections[].confidence	float	0.92	Model confidence score (0.0 - 1.0)
results[].detections[].x1, y1, x2, y2	int	120, 80, 200, 160	Bounding box coordinates (pixels)
results[].detections[].width, height	int	80, 80	Bounding box dimensions (pixels)

Key	Type	Example	Description
results[].filename	string	xray_001.jpg	Original filename of the processed image
results[].annotated_image_b64	string	iVBORw0KGgo...	Base64-encoded annotated X-ray image with bounding boxes drawn
results[].total_detections	integer	3	Number of detections found in this image
success	boolean	True	Overall success status
total_images	integer	2	Total number of images processed
total_detections	integer	5	Sum of all detections across all images

EXAMPLE RESPONSE

```
{
  "results": [
    {
      "detections": [
        {
          "class_name": "cavity",
          "confidence": 0.92,
          "x1": 120,
          "y1": 80,
          "x2": 200,
          "y2": 160,
          "width": 80,
          "height": 80
        },
        {
          "class_name": "filling",
          "confidence": 0.87,
          "x1": 300,
          "y1": 100,
          "x2": 370,
          "y2": 170,
          "width": 70,
          "height": 70
        }
      ],
      "filename": "xray_001.jpg",
      "annotated_image_b64": "iVBORw0KGgoAAAANSUHEug...(truncated)",
      "total_detections": 2
    }
  ],
  "success": true,
  "total_images": 1,
  "total_detections": 2
}
```

ENDPOINT 5 OF 7

POST

/api/correlate

Correlate Symptoms with X-ray Findings

Matches patient-reported symptoms (from the chatbot interview) with YOLO detection results. Produces natural-language explanations linking complaints to specific findings, sorted by match strength and severity.

REQUEST BODY (CorrelationRequest)

Key	Type	Example	Description
symptoms	dict	<code>{"has_pain":true,"pain_locatio...</code>	Patient symptom data (same structure as SymptomData fields)
detections	list[dict]	<code>[{"class_name":"cavity","confi...</code>	YOLO detection results array
image_width	integer	<code>800</code>	Width of the original X-ray image in pixels (for quadrant mapping)
image_height	integer	<code>600</code>	Height of the original X-ray image in pixels (for quadrant mapping)

EXAMPLE REQUEST

```
{
  "symptoms": {
    "has_pain": true,
    "pain_location": "lower-left",
    "pain_type": "throbbing",
    "has_swelling": true,
    "swelling_severity": "moderate"
  },
  "detections": [
    {
      "class_name": "cavity",
      "confidence": 0.92,
      "x1": 120,
      "y1": 80,
      "x2": 200,
      "y2": 160
    }
  ],
  "image_width": 800,
  "image_height": 600
}
```

RESPONSE BODY (JSON)

Key	Type	Example	Description
correlations	list[dict]	<code>[{...}]</code>	Array of correlation objects sorted by match strength

Key	Type	Example	Description
correlations[].detection_index	integer	0	Index of the detection in the original detections array
correlations[].class_name	string	cavity	YOLO detection class name
correlations[].clinical_name	string	Dental Caries (Cavity)	Human-readable clinical term
correlations[].confidence	float	0.92	Detection confidence score
correlations[].confidence_label	string	high confidence	Human-readable confidence label
correlations[].detection_region	string	Lower Left (Quadrant 3, teeth ...	Mapped dental quadrant of detection
correlations[].location_match	boolean	True	Whether symptom location matches detection quadrant
correlations[].match_type	string	strong	Match strength: strong swelling location additional
correlations[].severity	string	high	Severity level: high moderate low
correlations[].explanation	string	You mentioned throbbing pain i...	Natural-language correlation explanation
correlations[].urgency	string	Prompt treatment recommended t...	Urgency recommendation
unmatched_symptoms	list[str]	["You reported sharp pain in u...	Symptoms that could not be matched to any detection
total_detections	integer	3	Total number of YOLO detections processed
total_correlations	integer	2	Number of correlations with location match

EXAMPLE RESPONSE

```
{
  "correlations": [
    {
      "detection_index": 0,
      "class_name": "cavity",
      "clinical_name": "Dental Caries (Cavity)",
      "confidence": 0.92,
      "confidence_label": "high confidence",
      "detection_region": "Lower Left (Quadrant 3, teeth 31-38)",
      "location_match": true,
      "match_type": "strong",
      "severity": "high",
      "explanation": "You mentioned throbbing pain in the lower-left region. Our AI detected tooth decay that has damaged the tooth structure in the same area (92% confidence). The cavity may have reached the tooth nerve (pulp), causing inflammation (pulpitis) and throbbing pain.",
      "urgency": "Prompt treatment recommended to prevent further nerve damage."
    }
  ],
  "unmatched_symptoms": [],
  "total_detections": 1,
  "total_correlations": 1
}
```

```
} 
```

ENDPOINT 6 OF 7

POST /api/treatment-plan

Generate Treatment Recommendations

Takes YOLO detection results and produces structured treatment recommendations using a rule-based expert system. Maps each finding to a treatment plan, severity, and specialist referral. Accepts either a full API response or a raw detections list.

REQUEST BODY (TreatmentRequest)

Key	Type	Example	Description
api_response	dict null	{"results":[{"detections":[.....	Full YOLO API response object (preferred). Mutually exclusive with 'detections'.
detections	list[dict] null	[{"class_name":"cavity","confi...	Raw detection list (alternative input). Used if api_response is null.

EXAMPLE REQUEST

```
{
  "detections": [
    {
      "class_name": "cavity",
      "confidence": 0.92,
      "width": 80,
      "height": 80
    },
    {
      "class_name": "filling",
      "confidence": 0.85,
      "width": 70,
      "height": 70
    }
  ]
}
```

RESPONSE BODY (JSON)

Key	Type	Example	Description
patient_file	string	xray_001.jpg	Filename of the analyzed image
total_detections	integer	3	Total number of detections
summary	dict	{"cavity":2,"filling":1,"impla...	Count of each detection class found
recommendations	list[dict]	[{...}]	Array of treatment recommendation objects
recommendations[].finding	string	Large/Deep Cavity	Detection finding label
recommendations[].confidence	string	92%	Formatted detection confidence
recommendations[].area	integer	6400	Bounding box area (width×height) in pixels²

Key	Type	Example	Description
recommendations[].severity	string	high	Severity level: high moderate low
recommendations[].treatment	string	Root Canal Treatment (Endodont...	Recommended treatment plan
recommendations[].specialist	string	Endodontist	Specialist referral type
recommendations[].icon	string	.	Visual icon for the finding type
specialists	list[str]	["Endodontist", "General Dentis...	Unique list of specialists to refer to
disclaimer	string	.. This is an AI-assisted prel...	Medical disclaimer text

EXAMPLE RESPONSE

```
{
  "patient_file": "xray_001.jpg",
  "total_detections": 3,
  "summary": {
    "cavity": 2,
    "filling": 1,
    "implant": 0,
    "impacted": 0
  },
  "recommendations": [
    {
      "finding": "Large/Deep Cavity",
      "confidence": "92%",
      "area": 6400,
      "severity": "high",
      "treatment": "Root Canal Treatment (Endodontics) followed by a Crown restoration. Vitality test recommended.",
      "specialist": "Endodontist",
      "icon": "\ud83e\udddb7"
    },
    {
      "finding": "Small/Medium Cavity",
      "confidence": "78%",
      "area": 3200,
      "severity": "moderate",
      "treatment": "Dental Restoration \u2014 Composite or Amalgam filling to restore tooth structure.",
      "specialist": "General Dentist",
      "icon": "\ud83d\udd27"
    },
    {
      "finding": "Existing Filling",
      "confidence": "85%",
      "area": 4900,
      "severity": "low",
      "treatment": "Routine monitoring. Check margins clinically and radiographically for secondary caries.",
      "specialist": "General Dentist",
      "icon": "\u2705"
    }
  ],
  "specialists": [
    "Endodontist",
```

```
    "General Dentist"
  ],
  "disclaimer": "\u26a0\u2014 This is an AI-assisted preliminary analysis. Final diagnosis and treatment plan
must be verified by a licensed dental professional."
}
```

ENDPOINT 7 OF 7

POST

/api/ai-report

Generate AI Clinical Report (Gemini-powered)

Sends dental X-ray image(s) or PDFs to an external Gemini-powered treatment recommendation API. Returns a full AI-generated clinical report with HTML-formatted analysis, detection counts by class, and an annotated image.

REQUEST BODY (File Upload)

Key	Type	Example	Description
images	List[UploadFile]	<binary file data>	One or more dental X-ray images (PNG, JPG) or PDF files

EXAMPLE REQUEST

```
# Using cURL:
curl -X POST http://localhost:7860/api/ai-report \
  -F "images=@dental_xray.jpg"

# Using Python (requests):
import requests

url = "http://localhost:7860/api/ai-report"
files = {"images": open("dental_xray.jpg", "rb")}
response = requests.post(url, files=files)
print(response.json())
```

RESPONSE BODY (JSON (proxied from external Treatment API))

Key	Type	Example	Description
results	dict	{"xray_001.jpg": {...}}	Dictionary keyed by filename with analysis results
results.<filename>.total	integer	3	Total detections for this image
results.<filename>.by_class	dict	{"cavity":2,"filling":1}	Detection counts grouped by class
results.<filename>.image_b64	string	iVBORw0KGgo...	Base64-encoded annotated result image
results.<filename>.report	string (HTML)	Clinical Chart Note <p...>	HTML-formatted AI clinical report generated by Gemini
results.<filename>.error	string null	None	Error message if analysis failed for this image, else null

EXAMPLE RESPONSE

```
{
  "results": {
    "xray_001.jpg": {
      "total": 3,
      "by_class": {
        "cavity": 2,
```



```
    "filling": 1
  },
  "image_b64": "iVBORw0KGgoAAAANSUhEUg...(truncated)",
  "report": "<h2>Clinical Chart Note</h2><p><strong>Patient:</strong> Unknown</p><p><strong>Chief
Complaint:</strong> Dental evaluation</p><h3>Radiographic Findings</h3><ul><li>Two carious lesions
detected</li><li>One existing restoration noted</li></ul><h3>Assessment</h3><p>Moderate dental pathology
requiring intervention.</p><h3>Treatment Plan</h3><ol><li>Restore carious lesions with composite
resin</li><li>Monitor existing filling at next recall</li></ol>",
  "error": null
}
}
```

Appendix: Error Responses

Status	Meaning	Cause	Example Body
400	Bad Request	Missing or invalid input data	<code>{"error": "No valid images found in the uploaded files."}</code>
400	Bad Request	No detection input provided	<code>{"error": "Provide 'api_response' or 'detections'."}</code>
500	Internal Server Error	Unexpected processing error	<code>{"error": "Error processing X-ray: <details>"}</code>
502	Bad Gateway	External API returned non-200	<code>{"error": "Treatment API returned 502", "detail": "<response text>"}</code>
504	Gateway Timeout	External API timed out after retries	<code>{"error": "YOLO API timed out after retries. Please try again."}</code>
504	Gateway Timeout	AI report generation timed out	<code>{"error": "AI report timed out. Gemini may be loading · try again."}</code>

Appendix: External APIs

API	Method	URL	Request Key	Response
YOLO Detection	POST	<code>https://0xker-dental-x-ray-detection.hf.space/api/predict</code>	image (multipart)	JSON with detections + annotated image (b64)
YOLO Annotated	POST	<code>...predict?format=image</code>	image (multipart)	Raw annotated image bytes
Treatment / AI	POST	<code>https://0xker-treat-recommend.hf.space/api/predict</code>	image + model (multipart)	JSON with report (HTML), by_class, image_b64